

Arithmetic Progressions in Gap Sets of Numerical Semigroups

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Abstract

For a numerical semigroup S , we study arithmetic progressions contained in its gap set and let $\text{AP}(S)$ denote their maximal length. Standard Apéry theory gives $\text{AP}_d(S) = \lfloor F/d \rfloor + 1$ when the common difference d belongs to S , where F is the Frobenius number. This isolates differences in the gap set, which lead to a divisor-orbit Run Lemma in Kunz coordinates whose positional term yields an anchoring phenomenon. For the extremal function

$$M(A) = \max\{g(S) : \text{AP}(S) \leq A\},$$

we prove $M(3) = 7$ and $M(9) = 54$, and computer-assisted enumeration gives $M(8) = 48$. A prime-multiplicity construction gives the lower bound $A(P(A+1)-1)$, where $P(x)$ is the largest prime not exceeding x , while composite multiplicities incur explicit divisor-orbit losses. This leads to the prime-staircase conjecture $M(A) = A(P(A+1)-1)$ for $A \neq 3$.

The prime-staircase conjecture remains open in full. We verify it computer-assistively for $A \leq 4 \cdot 10^{18}$, $A \neq 3$, prove it unconditionally on a set of integers A of natural density one, and prove its eventual validity under a square-root-scale hypothesis on consecutive prime gaps. Known prime-gap estimates also yield $M(A) = (1 + o(1))A^2$. Finally, we obtain genus-realization results below the prime-staircase lower bound and corresponding information about the inverse extremal function. Accordingly, known information on consecutive prime gaps provides arithmetic input for the semigroup extremal problem. The principal remaining obstacle is structural: propagating divisor-orbit loss through the Kunz inequalities for the residual composite multiplicities.

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1 Introduction

A numerical semigroup is a cofinite additive submonoid $S \subseteq \mathbb{N}$, where $0 \in \mathbb{N}$. Its finite complement $G(S) = \mathbb{N} \setminus S$ is the gap set [13]. Classical invariants measure the number and location of these gaps: the genus $g = |G(S)|$, the Frobenius number $F = \max G(S)$,

the multiplicity $m = \min(S \setminus \{0\})$, and the Apéry sets. In this paper we study a different aspect of the gap set, namely additive patterns internal to it. For each $d \geq 1$, we define

$$\text{AP}_d(S) = \max\{k : \text{some } x > 0 \text{ satisfies } x, x + d, \dots, x + (k - 1)d \in G(S)\}.$$

We also set

$$\text{AP}(S) = \max_{d \geq 1} \text{AP}_d(S), \quad \Sigma(S) = \{(d, k) \in \mathbb{N}_{>0}^2 : \text{AP}_d(S) \geq k\}.$$

The internal combinatorics of numerical-semigroup gap sets has been studied from several complementary perspectives. Eliahou and Fromentin's gapset program treats $G(S)$ as a combinatorial object in its own right, using gapsets, filtrations, multiplicity, and related invariants [6, 7]. The consecutive- difference case belongs to the classical theory of gap blocks, also called deserts: $\text{AP}_1(S)$ is the length of the longest such block [12]. Residue-class statistics of gaps are naturally expressed through Apéry sets: Shor studies equidistribution modulo an integer and, more recently, parity distributions for free numerical semigroups [15, 16]. Kunz coordinates also admit a rich polyhedral interpretation through the Kunz cone, Kunz posets, and the Kunz fan [10, 4], with related genus-counting applications [3].

Rather than counting gaps or studying only their distribution among residue classes, we study arithmetic progressions contained in the gap set itself. Shor's questions concern residue statistics; our residue-prefix results lie in the same Apéry-set framework, but we ask for the longest ordered patterns supported by those classes and, for divisors of the multiplicity, by their cyclic interaction. The special case $d = 1$, corresponding to maximal blocks of consecutive gaps, is classical; in particular, its maximum length is $m - 1$ [12]. To our knowledge, the common-difference invariants $\text{AP}_d(S)$ for arbitrary d , the global invariant $\text{AP}(S)$, and the extremal function $M(A)$ have not previously been studied in this form.

The analysis rests on the following structural dichotomy:

$$d \in S \quad \text{versus} \quad d \in G(S).$$

Subtraction by an element of S preserves gaps while the result remains positive. Consequently, Theorem 2.5 shows that, for $d \in S_{>0}$, the Apéry classes determine the entire distribution of gaps modulo d and give

$$\text{AP}_d(S) = \left\lfloor \frac{F}{d} \right\rfloor + 1. \quad (1)$$

This identity is a standard consequence of Apéry-set theory. Together with the classical consecutive-gap case, it gives the hierarchy

$$\begin{aligned} d = 1 & : \text{AP}_1(S) = m - 1, & \text{classical gap blocks or deserts;} \\ d \in S_{>0} & : \text{AP}_d(S) = \left\lfloor \frac{F}{d} \right\rfloor + 1, & \text{standard Apéry structure;} \\ d \in G(S) \setminus \{1\} & : \text{the genuinely difficult part of the AP spectrum.} \end{aligned}$$

We record the semigroup-difference formulation because it isolates this last component.

The global invariant already satisfies a useful universal bound:

$$\text{AP}(S) \geq \max \left\{ m - 1, \left\lceil \frac{g}{m - 1} \right\rceil \right\} \geq \lceil \sqrt{g} \rceil. \quad (2)$$

The main semigroup mechanism is the divisor-orbit formula of Lemma 3.1. In Kunz coordinates, we write

$$\text{Ap}(S, m) = \{0, w_1, \dots, w_{m-1}\}, \quad w_i = i + mk_i.$$

From the polyhedral viewpoint, the Run Lemma imposes divisor-orbit constraints on Kunz vectors; its positional term retains information not visible in coordinate-wise height bounds.

For $d \mid m$, a nonzero orbit under addition by d reads the gap-prefixes of its residue classes cyclically. If its length is $\ell = m/d$, its minimum Kunz height is μ , and the first minimum occurs at zero-indexed position ρ , then the longest gap run on that orbit has length

$$\ell\mu + \rho. \quad (3)$$

The term ρ is essential. Under an upper bound on $\text{AP}(S)$, an extremal value of μ can force the minimum to occur at a prescribed position. We refer to this positional restriction as *anchoring*. The Run Lemma and anchoring are the source of both the small exact computations and the general penalty for composite multiplicities.

These constraints motivate the extremal function and its inverse problem, which we define by

$$M(A) = \max\{g(S) : \text{AP}(S) \leq A\}, \quad a(g) = \min\{\text{AP}(S) : g(S) = g\}.$$

The estimate (2) gives $M(A) \leq A^2$, but it does not detect the arithmetic restrictions imposed by differences in $G(S)$.

The first nontrivial extremal values are

$$M(3) = 7, \quad M(8) = 48, \quad M(9) = 54.$$

Here $M(8) = 48$ is presently computer-assisted, while Theorem 4.5 proves $M(9) = 54$ by combining divisor-orbit constraints, Kunz propagation, and a finite Kunz certificate in the remaining multiplicities. An explicit multiplicity-9 Kunz vector

$$(3, 3, 6, 6, 9, 9, 9, 9)$$

attains genus 54 and has $\text{AP}(S) = 9$. This is the first plateau where the bound $M(A) \leq A^2$ is far from the true value.

The general construction is controlled by primes. Let

$$P(x) = \max\{p \leq x : p \text{ is prime}\}.$$

For a prime $p \leq A + 1$, the uniform multiplicity- p Kunz vector $(A, \dots, A)$ has genus $A(p - 1)$ and AP length A . This construction gives

$$M(A) \geq A(P(A + 1) - 1). \quad (4)$$

The resulting *prime staircase conjecture* predicts equality for every $A \neq 3$, while $M(3) = 7$ is exceptional. No uniqueness assertion is part of the conjecture. Indeed, at $A = 9$ the maximum is attained by non-isomorphic semigroups of multiplicities 7 and 9. Determining the structure and uniqueness of maximizers is therefore a separate problem from determining the value of $M(A)$.

The opposite force comes from compositeness. If a prime q divides the multiplicity m , the Run Lemma gives the genus loss

$$g \leq A(m-1) - \left(\frac{m}{q} - 1\right) \left(A - \left\lfloor \frac{A}{q} \right\rfloor\right). \quad (5)$$

For the least prime factor this loss is greater than $A(\sqrt{m} - 2)$. Comparing it with the extra capacity obtained by taking $m > P(A+1)$ is precisely where estimates for gaps between consecutive primes enter the semigroup problem.

The extremal problem is therefore governed by a competition between two effects: prime multiplicities attain the naive Kunz capacity, whereas composite multiplicities are subject to divisor-orbit losses. If $p = P(A+1)$, the backward prime distance $A+1-p$ measures how much additional multiplicity a composite competitor can have beyond the largest available prime multiplicity. This is where results on gaps between consecutive primes enter the argument.

The number-theoretic input is entirely one-directional: we use known estimates for prime gaps to obtain consequences for the semigroup extremal problem, and we obtain no new bounds or other results concerning prime gaps. The relevant input concerns upper bounds for gaps between consecutive primes, or equivalently primes in short intervals, rather than the bounded-gap problem. Pointwise, we use the Baker–Harman–Pintz exponent 0.525. For density statements we use Stadlmann’s mean-square estimate for consecutive prime gaps; Heath-Brown’s classical estimate gives a weaker historical comparison [2, 9, 19]. We also record conditional consequences of standard conjectural prime-gap bounds.

The full prime-staircase conjecture remains open. The partial results have three distinct scopes: the conjectural formula is verified computer-assistively for $1 \leq A \leq 4 \cdot 10^{18}$ with $A \neq 3$; it holds unconditionally for all but $O_\varepsilon(x^{73/100+\varepsilon})$ integers $A \leq x$; and, under the square-root gap hypothesis $g_p \leq \sqrt{p} - 1$, it holds for every sufficiently large A .

The principal unresolved step is semigroup-theoretic rather than a further application of the prime-gap machinery. Known prime-gap estimates allow the criterion above to eliminate most multiplicities, but the surviving prime-square and balanced-semiprime cases require additional loss propagated through Run-Lemma anchoring and Kunz sub-additivity. Establishing such a propagated-loss theorem, formulated as Problem 8.1, is the main structural route toward an unconditional proof of the full conjecture.

The prime construction also fills the genera below its upper endpoint. For every $A \geq 1$, Theorem 7.3 realizes each genus g satisfying

$$1 \leq g \leq A(P(A+1) - 1)$$

by a numerical semigroup with $\text{AP}(S) \leq A$. Every proved exact jump of the prime staircase therefore yields a corresponding plateau for $a(g)$.

Two further unconditional consequences sharpen the asymptotic picture. The first characterizes the square maximum:

$$M(A) = A^2 \iff A+1 \text{ is prime.}$$

The second concerns the deficit $\Delta(A) = A^2 - M(A)$ and gives

$$\Delta(A) \ll A^{1.525}, \quad M(A) = (1 + o(1))A^2.$$

Wherever the exact staircase formula is known, the deficit satisfies

$$\frac{\Delta(A)}{A} = A + 1 - P(A + 1),$$

so the normalized extremal deficit is precisely the backward prime distance.

Section 2 develops the elementary and Apéry-theoretic structure. Section 3 proves the Run Lemma and the composite-loss theorem. Section 4 treats the small extremal values and the plateau at nine. Sections 5 and 6 develop the prime staircase and derive consequences from known estimates for gaps between consecutive primes. Section 7 proves the realization and inverse-plateau results. Section 8 concludes with the remaining structural problems, and the appendix records the computational certificates.

2 Arithmetic progressions in numerical semigroup gaps

Unless stated otherwise, S has positive genus. We use the notation

$$G(S) = \mathbb{N} \setminus S, \quad g = |G(S)|, \quad F = \max G(S), \quad m = \min(S \setminus \{0\}).$$

When $S = \mathbb{N}$, the gap set is empty; one may set $\text{AP}_d(S) = \text{AP}(S) = 0$, but F is undefined and statements involving it do not apply.

Lemma 2.1 (Downward closure). *If $x \in G(S)$ and $s \in S$ satisfy $x - s > 0$, then $x - s \in G(S)$.*

Proof. Otherwise $x - s \in S$, and closure under addition gives $x = (x - s) + s \in S$, a contradiction. $\square$

Proposition 2.2 (Consecutive gaps). *For every numerical semigroup of positive genus and multiplicity m , we have*

$$\boxed{\text{AP}_1(S) = m - 1.}$$

Proof. The integers $1, 2, \dots, m - 1$ are gaps, so $\text{AP}_1(S) \geq m - 1$. On the other hand, any m consecutive integers contain a multiple of m . Since $m \in S$ and S is closed under addition, every nonnegative multiple of m belongs to S . Hence no m consecutive positive integers can all be gaps, and therefore $\text{AP}_1(S) \leq m - 1$. $\square$

This is the classical longest-gap-block observation; Moree uses the term *gap block* and notes the alternative terminology *desert* [12].

The consecutive case leads naturally to the more restrictive question of when the entire gap set forms a single arithmetic progression. There are only two possibilities.

Proposition 2.3. *Assume $g \geq 1$. The gap set $G(S)$ is a single arithmetic progression if and only if either*

1. $S = \{0, g + 1, g + 2, \dots\}$ and $G(S) = \{1, 2, \dots, g\}$; or
2. $S = \langle 2, 2g + 1 \rangle$ and $G(S) = \{1, 3, 5, \dots, 2g - 1\}$.

For $g = 1$, the two descriptions coincide.

Proof. Since $1 \in G(S)$, an arithmetic progression equal to $G(S)$ has the form

$$\{1, 1 + d, \dots, 1 + (g - 1)d\}.$$

If $d = 1$, the semigroup is ordinary. If $d = 2$, its gaps are precisely the odd integers below $2g + 1$, and the semigroup is $\langle 2, 2g + 1 \rangle$. If $d \geq 3$, then $2, 3 \in S$, and therefore $\langle 2, 3 \rangle = \mathbb{N} \setminus \{1\} \subseteq S$. It follows that $G(S) = \{1\}$, which is the common genus-one case. The converse follows directly from the two displayed gap sets. $\square$

The genus-zero semigroup $\mathbb{N}$ has empty gap set and is excluded from this classification.

For arbitrary common differences, the relevant structure is the distribution of the gaps among residue classes. Fix $d \in S_{>0}$ and recall that

$$\text{Ap}(S, d) = \{s \in S : s - d \notin S\} = \{w_0, w_1, \dots, w_{d-1}\},$$

where $w_r \equiv r \pmod{d}$ and $0 \leq r < d$. In particular, $w_0 = 0$.

Proposition 2.4. *For $1 \leq r \leq d - 1$, the gaps in residue class r are*

$$G(S) \cap (r + d\mathbb{Z}) = \{r, r + d, \dots, w_r - d\}.$$

Consequently, this residue class contains

$$\frac{w_r - r}{d} = \left\lfloor \frac{w_r}{d} \right\rfloor,$$

gaps. Taking the maximum over the nonzero residue classes gives

$$\text{AP}_d(S) = \max_{w \in \text{Ap}(S, d)} \left\lfloor \frac{w}{d} \right\rfloor.$$

Proof. The element w_r is the least member of S congruent to r modulo d . It follows that the positive integers $r, r + d, \dots, w_r - d$ are gaps, while $w_r + jd \in S$ for $j \geq 0$. Writing $w_r = r + a_r d$ shows that the displayed progression has $a_r = (w_r - r)/d = \lfloor w_r/d \rfloor$ terms. Taking the largest residue-class prefix gives the final formula. $\square$

Summing these prefix lengths gives Selmer's genus formula

$$g = \frac{1}{d} \sum_{w \in \text{Ap}(S, d)} w - \frac{d-1}{2}. \quad (6)$$

See [14]; standard background on numerical semigroups and Apéry sets is available in [13].

This is the Apéry residue-class framework discussed in the Introduction; see [15, 16] for related distributional questions. Here the prefix lengths will control arithmetic progressions.

When the common difference itself belongs to the semigroup, the residue-class formula depends only on F and d .

Theorem 2.5 (Semigroup differences). *For every $d \in S_{>0}$, we have*

$$\boxed{\text{AP}_d(S) = \left\lfloor \frac{F}{d} \right\rfloor + 1.}$$

Proof. The standard identity $\max \text{Ap}(S, d) = F + d$ and Proposition 2.4 give

$$\text{AP}_d(S) = \left\lfloor \frac{F + d}{d} \right\rfloor = \left\lfloor \frac{F}{d} \right\rfloor + 1.$$

For a direct description, write $F = cd + r$ with $0 \leq r < d$. Since $d \in S$ and S is closed under addition, every positive multiple of d belongs to S . As $F \notin S$, one cannot have $d \mid F$, and hence $1 \leq r \leq d - 1$. Repeated downward closure gives the gap progression

$$r, r + d, \dots, r + cd = F$$

of length $c + 1$. Conversely, if $x, x + d, \dots, x + (k - 1)d$ is a gap progression, then $x \geq 1$ and its last term is at most F . These inequalities give

$$1 + (k - 1)d \leq x + (k - 1)d \leq F,$$

so $k \leq \lfloor F/d \rfloor + 1$. □

Corollary 2.6 (Semigroup-difference spectrum). *The semigroup-difference part of the spectrum is*

$$\Sigma(S) \cap (S_{>0} \times \mathbb{N}_{>0}) = \{(d, k) : d \in S, (k - 1)d \leq F\}.$$

Remark 2.7. The Apéry set records the full residue-class distribution of gaps, whereas Theorem 2.5 shows that its longest prefix depends only on F and d . Therefore, the part of $\Sigma(S)$ with $d \in S$ is governed by classical data; apart from the classical case $d = 1$, the new component occurs for $d \in G(S) \setminus \{1\}$.

Although gap differences contain the new part of the spectrum, the classical residue prefixes already impose a useful lower bound on the global invariant.

Theorem 2.8. *For every numerical semigroup of positive genus, we have*

$$\boxed{\text{AP}(S) \geq \max \left\{ m - 1, \left\lceil \frac{g}{m - 1} \right\rceil \right\} .}$$

Proof. Proposition 2.2 gives $\text{AP}_1(S) = m - 1$. By Proposition 2.4 at $d = m$, the gaps split into $m - 1$ nonzero residue classes modulo m . Writing $k_i = (w_i - i)/m$ for their lengths gives $\sum_i k_i = g$ and

$$\text{AP}_m(S) = \max_i k_i \geq \left\lceil \frac{g}{m - 1} \right\rceil .$$

Combining these two bounds gives

$$\text{AP}(S) \geq \max\{\text{AP}_1(S), \text{AP}_m(S)\} \geq \max \left\{ m - 1, \left\lceil \frac{g}{m - 1} \right\rceil \right\} .$$

□

Corollary 2.9. *Every numerical semigroup of positive genus satisfies*

$$\boxed{\text{AP}(S) \geq \lceil \sqrt{g} \rceil .}$$

Proof. Apply $\max\{x, g/x\} \geq \sqrt{g}$ with $x = m - 1$ and use integrality. □

Combining Theorems 2.5 and 2.8 at $d = m$ gives

$$\text{AP}_m(S) = \left\lfloor \frac{F}{m} \right\rfloor + 1 \geq \left\lceil \frac{g}{m - 1} \right\rceil .$$

Consequently, the genus satisfies

$$g \leq (m - 1) \left(\left\lfloor \frac{F}{m} \right\rfloor + 1 \right) .$$

3 Kunz coordinates and divisor-orbit constraints

With respect to the multiplicity, we write

$$\text{Ap}(S, m) = \{0, w_1, \dots, w_{m-1}\}, \quad w_i = i + mk_i.$$

The positive integers k_i are the Kunz coordinates. The coordinate k_i is the number of gaps congruent to i modulo m , so

$$g = \sum_{i=1}^{m-1} k_i. \quad (7)$$

For $1 \leq i, j \leq m-1$, the Kunz coordinates satisfy

$$k_i + k_j \geq k_{i+j} \quad (i+j < m), \quad (8)$$

$$k_i + k_j + 1 \geq k_{i+j-m} \quad (i+j > m). \quad (9)$$

When $i+j = m$, no nontrivial coordinate inequality is required. Conversely, every positive integral vector satisfying these inequalities is the Kunz vector of a numerical semigroup of multiplicity m ; see [13]. The polyhedral background appears in [10, 4], with related genus-counting applications in [3].

If $\text{AP}(S) \leq A$, then Proposition 2.4 at $d = m$ gives

$$1 \leq k_i \leq A \quad (1 \leq i < m). \quad (10)$$

The coordinatewise box (10) does not record how a progression moves cyclically through residue classes. When its common difference divides the multiplicity, this interaction is exact.

Lemma 3.1 (Run Lemma). *Let $d \mid m$ with $1 \leq d < m$, and let $1 \leq r \leq d-1$. In cyclic order, the orbit of r under addition by d modulo m is*

$$c_0 = r, c_1 = r + d, \dots, c_{\ell-1} = r + m - d, \quad \ell = \frac{m}{d}.$$

Define the minimum height and its first position by

$$\mu = \min_{0 \leq j < \ell} k_{c_j}, \quad \rho = \min\{j : k_{c_j} = \mu\}.$$

The longest gap progression of difference d contained in this orbit then has length

$$\boxed{\text{AP}_d(S)|_{\mathcal{O}_r} = \ell\mu + \rho = \frac{m}{d}\mu + \rho.}$$

Proof. We read the positive integers in the orbit with step d :

$$r, r + d, r + 2d, \dots$$

If $n = h\ell + j$ with $0 \leq j < \ell$, then $r + nd = c_j + hm$. The class c_j contributes the gap-prefix

$$c_j, c_j + m, \dots, c_j + (k_{c_j} - 1)m.$$

In this way, the orbit is read cyclically, one complete cycle at each height level h . At every level $h < \mu$, all ℓ entries are gaps, producing $\ell\mu$ initial gaps. At level $h = \mu$, the first ρ classes still contribute gaps, while $c_\rho + \mu m$ is the first nongap. The initial run has length $\ell\mu + \rho$. At every later level, the class c_ρ again contributes a nongap, so every later run has length at most $\ell - 1$. Since $\mu \geq 1$, the initial run is strictly longer. $\square$

The positional term ρ strengthens the minimum-height constraint whenever the latter is close to its largest possible value.

Corollary 3.2 (Orbit constraint). *If $\text{AP}(S) \leq A$, then every nonzero divisor orbit in Lemma 3.1 satisfies*

$$\boxed{\ell\mu + \rho \leq A.}$$

In particular, $\mu \leq \lfloor A/\ell \rfloor$; if equality holds, the first minimum is restricted by

$$\rho \leq A - \ell \left\lfloor \frac{A}{\ell} \right\rfloor.$$

Remark 3.3 (Anchoring). The position ρ contains information absent from the minimum-height bound. When μ is extremal, the first minimum can be forced to the beginning of the cyclic order. This anchoring phenomenon is decisive in the plateau calculation of Section 4.

Summing the orbit constraints converts this local information into a global genus loss for composite multiplicities.

Theorem 3.4 (Composite-multiplicity loss bound). *Let $\text{AP}(S) \leq A$, let m be the multiplicity, and let $q \mid m$ be prime. Under these assumptions, the genus satisfies*

$$\boxed{g \leq \left(\frac{m}{q} - 1\right) \left((q-1)A + \left\lfloor \frac{A}{q} \right\rfloor\right) + (q-1)A.}$$

Equivalently, the bound can be written as

$$\boxed{g \leq A(m-1) - \left(\frac{m}{q} - 1\right) \left(A - \left\lfloor \frac{A}{q} \right\rfloor\right).} \quad (11)$$

Proof. We set $d = m/q$. Addition by d modulo m has m/q cycles of length q . One contains residue 0; the other $m/q - 1$ cycles contain only nonzero residues. On each nonzero cycle, Corollary 3.2 gives $\mu \leq \lfloor A/q \rfloor$. Its coordinate sum is therefore at most

$$(q-1)A + \left\lfloor \frac{A}{q} \right\rfloor.$$

The zero cycle contains $q-1$ nonzero coordinates and contributes at most $(q-1)A$. Adding the contributions of all cycles gives

$$\begin{aligned} g &\leq \left(\frac{m}{q} - 1\right) \left((q-1)A + \left\lfloor \frac{A}{q} \right\rfloor\right) + (q-1)A \\ &= A(m-1) - \left(\frac{m}{q} - 1\right) \left(A - \left\lfloor \frac{A}{q} \right\rfloor\right), \end{aligned}$$

which is (11). □

Remark 3.5. The term $A(m-1)$ is the naive genus capacity from (10). The subtracted term in (11) is the genus penalty forced by the divisor orbits.

4 Small extremal values and the plateau at nine

The composite-multiplicity loss already determines the exceptional value at $A = 3$.

Applying Theorem 3.4 with $(A, m, q) = (3, 4, 2)$ gives

$$g \leq 3(4 - 1) - (2 - 1) \left(3 - \left\lfloor \frac{3}{2} \right\rfloor \right) = 7.$$

For $m \leq 3$, we have $g \leq 3(m - 1) \leq 6$. To attain the remaining upper bound, consider the semigroup

$$S = \langle 4, 7, 13 \rangle,$$

which has Kunz vector $(3, 3, 1)$ and gap set

$$G(S) = \{1, 2, 3, 5, 6, 9, 10\}.$$

For differences 1, 2, 3, 4, the following progressions have length three, respectively:

$$(1, 2, 3), \quad (1, 3, 5), \quad (3, 6, 9), \quad (1, 5, 9).$$

A direct check of the displayed gap set shows that no progression with any of these four differences has four terms. If $d \geq 5$, a four-term progression has span $3d > F - 1 = 9$. We conclude that $\text{AP}(S) = 3$, and hence

$$\boxed{M(3) = 7.} \tag{12}$$

At $A = 8$, a direct exhaustive search gives the exact extremal genus.

Computer-assisted theorem 4.1 (The value at eight). *The extremal value at $A = 8$ is*

$$\boxed{M(8) = 48.}$$

Computer-assisted proof. The accompanying directory `computational_reproducibility_files` contains the program `direct_ap_search.c`, which first verifies the prime-multiplicity lower-bound example directly. The naive coordinate bound then excludes multiplicities at or below that prime, and the program performs exhaustive branch-and-bound over the remaining multiplicities $m \leq 9$ and coordinate vectors with $1 \leq k_i \leq 8$. It tests the Kunz inequalities and prunes only when the largest possible genus on a branch cannot exceed the verified lower bound. For every remaining candidate, it constructs the finite gap set, computes gap arithmetic progressions directly, and maximizes $\sum_i k_i$. It uses no Run-Lemma constraint and returns $M(8) = 48$. The corresponding output is designated `direct_ap_A8.log` in the same directory. The maximum is attained at prime multiplicity 7 by the construction of Proposition 5.1 below. $\square$

For comparison, Theorem 3.4 with $(A, m, q) = (8, 8, 2)$ gives the stronger multiplicity-specific bound $g \leq 44$.

At $A = 9$, the single-divisor estimate alone does not recover the sharp bound. Assume that $\text{AP}(S) \leq 9$; the Kunz coordinates then satisfy $1 \leq k_i \leq 9$.

For $m = 10$ and $d = 5$, the relevant pairs are (k_r, k_{r+5}) , where $1 \leq r \leq 4$. The Run Lemma gives

$$\min(k_r, k_{r+5}) \leq 4. \tag{13}$$

For $d = 2$, the odd orbit $(1, 3, 5, 7, 9)$ has length 5. The corresponding orbit constraint is therefore

$$\min(k_1, k_3, k_5, k_7, k_9) = 1. \quad (14)$$

For $m = 9$ and $d = 3$, the two nonzero cycles are $(1, 4, 7)$ and $(2, 5, 8)$. On either cycle, we have $3\mu + \rho \leq 9$. This inequality implies that $\mu \leq 3$ and, if $\mu = 3$, that $\rho = 0$. More explicitly, it gives

$$\begin{aligned} \min(k_1, k_4, k_7) = 3 &\implies k_1 = 3, \\ \min(k_2, k_5, k_8) = 3 &\implies k_2 = 3. \end{aligned} \quad (15)$$

For $m = 8$ and $d = 4$, the pair constraints are

$$\min(k_i, k_{i+4}) \leq 4 \quad (1 \leq i \leq 3). \quad (16)$$

The $d = 2$ odd orbit $(1, 3, 5, 7)$ also satisfies

$$4\mu + \rho \leq 9. \quad (17)$$

Lemma 4.2. *If $\text{AP}(S) \leq 9$ and $m \leq 7$, then the genus satisfies $g \leq 54$.*

Proof. The coordinate bound gives $g = \sum_{i=1}^{m-1} k_i \leq 9(m-1) \leq 54$. $\square$

Proposition 4.3. *If $\text{AP}(S) \leq 9$ and $m = 8$, then the genus satisfies $g \leq 48$.*

Proof. Theorem 3.4 with $q = 2$ gives

$$g \leq 9(8-1) - (4-1) \left(9 - \left\lfloor \frac{9}{2} \right\rfloor \right) = 48.$$

Equivalently, each pair in (16) contributes at most 13, and $k_4 \leq 9$. $\square$

For the remaining multiplicities, anchoring interacts with Kunz subadditivity. For instance, if $m = 9$ and $k_1 = k_2 = 3$, then

$$k_3 \leq k_1 + k_2 = 6, \quad k_4 \leq 2k_2 = 6,$$

so $g \leq 3 + 3 + 6 + 6 + 9 + 9 + 9 + 9 = 54$. The doubling map on nonzero residues contains the cycle

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 7 \rightarrow 5 \rightarrow 1 \pmod{9},$$

which propagates such bounds through the remaining coordinates.

Proposition 4.4 (Finite Kunz certification). *For $m = 8, 9, 10$, consider the maximum of $\sum_i k_i$ over vectors in $\{1, \dots, 9\}^{m-1}$ satisfying all Kunz inequalities and the orbit constraints (13)–(17). Exhaustive enumeration gives*

m	$\max \sum_i k_i$	unique maximizing vector
8	44	$(7, 9, 2, 9, 4, 4, 9)$
9	54	$(3, 3, 6, 6, 9, 9, 9, 9)$
10	53	$(9, 9, 9, 8, 6, 4, 4, 3, 1)$.

Computer-assisted proof. The finite search checks only $1 \leq k_i \leq 9$, the Kunz inequalities, and the displayed Run-Lemma consequences. It neither constructs gap sets nor computes any $\text{AP}_d(S)$. It was performed by the program `plateau_certificate.c` in the accompanying directory `computational_reproducibility_files`; the corresponding output is designated `plateau_certificate.log`. An independent reordered branch-and-bound implementation is provided in `plateau_certificate_independent.c`, with output in `plateau_certificate_independent.log`. The constraints are necessary for $\text{AP}(S) \leq 9$; no converse characterization is asserted. They suffice for the stated optimization. $\square$

The preceding bounds cover every possible multiplicity and therefore give the desired plateau estimate.

Theorem 4.5 (Plateau at nine). *If $\text{AP}(S) \leq 9$, then the genus satisfies*

$$g(S) \leq 54.$$

Proof. Theorem 2.8 gives $m - 1 \leq 9$, so $m \leq 10$. Lemma 4.2 handles $m \leq 7$, Proposition 4.3 handles $m = 8$, and Proposition 4.4 gives $g \leq 54$ for $m = 9$ and $g \leq 53$ for $m = 10$. $\square$

Consider the multiplicity-9 Kunz vector

$$(k_1, \dots, k_8) = (3, 3, 6, 6, 9, 9, 9, 9). \quad (18)$$

It satisfies the Kunz inequalities. Indeed, the minimum left sides of the unwrapped inequalities with targets $2, \dots, 8$ are $6, 6, 6, 9, 9, 12, 12$, while those of the wrapped inequalities with targets $1, \dots, 7$ are $13, 16, 16, 19, 19, 19, 19$. Its genus is 54, and

$$F = \max_{1 \leq i \leq 8} (i + 9k_i) - 9 = 80.$$

Proposition 4.6. *The semigroup defined by (18) satisfies $\text{AP}(S) = 9$.*

Proof. Suppose first that $\gcd(d, 9) = 1$. Every nine consecutive terms of a d -progression represent all residues modulo 9, so one is a positive multiple of 9 and belongs to S . Therefore, $\text{AP}_d(S) \leq 8$.

For $d = 3$, the nonzero height triples are $(3, 6, 9)$ and $(3, 9, 9)$. The Run Lemma gives length $3 \cdot 3 = 9$ on each; the zero orbit has a multiple of 9 every third term. It follows that $\text{AP}_3(S) = 9$.

For $d = 6$, direct inspection of the residue prefixes gives the following longest-run lengths:

$x \bmod 6$	1	2	3	4	5	0
length	6	6	2	4	4	2.

For example, the two length-six runs are $(1, 7, 13, 19, 25, 31)$ and $(2, 8, 14, 20, 26, 32)$; their next terms are nongaps. This is a direct inspection, since $6 \nmid 9$.

If $\gcd(d, 9) = 3$ and $d \notin \{3, 6\}$, then $d \geq 12$, and every gap lies in $[1, 80]$. Therefore, the progression has length at most $\lceil 79/12 \rceil + 1 = 7$. Finally, if $9 \mid d$, the progression remains in one residue class, containing at most 9 gaps. Equality occurs for $d = 9$. These cases cover all $d > 0$. $\square$

Combining the upper bound with the sharpness construction gives

$$\boxed{M(9) = 54.} \tag{19}$$

The estimate $M(9) \leq 81$ from Corollary 2.9 misses the loss imposed by gap differences; the plateau calculation exhibits the role of anchoring and Kunz propagation.

5 The prime staircase and composite multiplicities

For $x \geq 2$, we denote by $P(x)$ the largest prime not exceeding x .

Proposition 5.1 (Prime staircase construction). *Let $p \leq A+1$ be prime. The semigroup with multiplicity p and uniform Kunz vector $(A, \dots, A)$ satisfies*

$$\text{AP}(S) = A, \quad g = A(p-1).$$

Proof. The uniform vector satisfies all Kunz inequalities. If $p \nmid d$, every p consecutive terms of a d -progression contain a positive multiple of p , so $\text{AP}_d(S) \leq p-1 \leq A$. If $p \mid d$, the progression stays in one residue class, and each nonzero class contains exactly A gaps. Hence $\text{AP}_d(S) \leq A$, with equality for $d = p$. $\square$

Corollary 5.2 (Prime staircase lower bound). *For every $A \geq 1$, we have*

$$\boxed{M(A) \geq A(P(A+1) - 1).}$$

Theorem 5.3 (Quadratic growth). *For every $A \geq 1$, the extremal function satisfies*

$$\frac{A(A-1)}{2} \leq M(A) \leq A^2.$$

In particular, $M(A) = \Theta(A^2)$.

Proof. The upper bound is Corollary 2.9. Bertrand's postulate gives $P(A+1) > (A+1)/2$ for $A \geq 2$, and the lower estimate follows from Corollary 5.2; the case $A = 1$ follows directly. $\square$

The prime construction and the quadratic upper bound suggest that the preceding lower bound is usually sharp.

Conjecture 5.4 (Prime staircase conjecture). *For every $A \geq 1$ with $A \neq 3$, the conjectural formula is*

$$\boxed{M(A) = A(P(A+1) - 1).}$$

At the exceptional value, $M(3) = 7$.

Direct computation for the first values gives the following comparison.

A	1	2	3	4	5	6	7	8	9	10	11	12
$M(A)$	1	4	7	16	20	36	42	48	54	100	110	144
$A(P(A+1) - 1)$	1	4	6	16	20	36	42	48	54	100	110	144.

The rows agree for $1 \leq A \leq 12$ except at $A = 3$, the exceptional value established in (12). Therefore, from $A = 4$ through 12, the prime staircase exactly predicts the directly computed extremal genus. This is computational evidence, not a proof of the conjecture, and is independent of the divisor-orbit criterion used below.

The conjecture does not assert uniqueness. At $A = 9$, both the uniform multiplicity-7 semigroup and the multiplicity-9 semigroup in (18) are maximizers.

Proposition 5.5 (Composite-multiplicity reduction). *Let $p = P(A + 1)$. To prove the conjectured upper bound, it suffices to consider*

$$\boxed{p < m \leq A + 1.}$$

Every such multiplicity is composite.

Proof. Theorem 2.8 gives $m \leq A + 1$. If $m \leq p$, then $g \leq A(m - 1) \leq A(p - 1)$. By definition of p , there is no prime r satisfying $p < r \leq A + 1$. $\square$

For a prime divisor $q \mid m$, Theorem 3.4 proves the desired upper bound whenever

$$\left(\frac{m}{q} - 1\right) \left(A - \left\lfloor \frac{A}{q} \right\rfloor\right) \geq A(m - p). \quad (20)$$

The left side is the Run-Lemma loss, and the right side is the excess capacity obtained by using $m > p$.

To compare these two quantities uniformly, we use the least prime factor of the competing multiplicity.

Lemma 5.6 (Least-factor loss bound). *Let m be composite and q_0 its least prime factor. The associated loss satisfies*

$$\boxed{\left(\frac{m}{q_0} - 1\right) \left(A - \left\lfloor \frac{A}{q_0} \right\rfloor\right) \geq A \frac{(\sqrt{m} - 1)^2}{\sqrt{m}} > A(\sqrt{m} - 2).}$$

Proof. We first observe that

$$A - \left\lfloor \frac{A}{q_0} \right\rfloor \geq A \left(1 - \frac{1}{q_0}\right).$$

To analyze the remaining factor, define

$$h(t) = \left(\frac{m}{t} - 1\right) \left(1 - \frac{1}{t}\right).$$

For this function, differentiation gives

$$h'(t) = \frac{2m - (m + 1)t}{t^3} < 0 \quad (t \geq 2).$$

Since $q_0 \leq \sqrt{m}$ and h is decreasing on the relevant interval, we obtain

$$h(q_0) \geq h(\sqrt{m}) = \frac{(\sqrt{m} - 1)^2}{\sqrt{m}} = \sqrt{m} - 2 + \frac{1}{\sqrt{m}} > \sqrt{m} - 2.$$

Multiplying by A proves the result. $\square$

The least-factor estimate rules out every composite competitor once the backward distance to the preceding prime is sufficiently small.

Theorem 5.7 (Prime-gap criterion). *Let $p = P(A + 1)$, and assume that*

$$\boxed{A + 1 - p \leq \sqrt{p} - 2}. \quad (21)$$

Under this assumption, we have

$$\boxed{M(A) = A(p - 1).}$$

Proof. By Proposition 5.5, only composite $p < m \leq A + 1$ require consideration. Theorem 3.4 and Lemma 5.6 give

$$g < A(m - 1) - A(\sqrt{m} - 2).$$

Since $m > p$ and (21) holds, we have

$$m - p \leq A + 1 - p \leq \sqrt{p} - 2 < \sqrt{m} - 2.$$

Combining these inequalities yields $g < A(m - 1) - A(m - p) = A(p - 1)$. The prime staircase construction attains $A(p - 1)$. $\square$

The theorem identifies the threshold at which prime-gap estimates imply optimality of the prime construction: the backward distance from $A + 1$ to its preceding prime must be at most the square-root threshold. A single divisor does not always suffice. At $A = 9$, $p = 7$, and

$$g \leq 60 \quad (m = 9, q = 3), \quad g \leq 61 \quad (m = 10, q = 2),$$

whereas the sharp value is 54. The additional anchoring and Kunz propagation of Section 4 are genuinely stronger.

6 Consequences of prime-gap estimates

Known information on consecutive prime gaps enters the extremal problem through Theorem 5.7. It yields three statements with distinct statuses: finite computer-assisted verification, unconditional density-one validity, and conditional eventual validity.

Proposition 6.1 (Finite verification of the composite criterion). *For $1 \leq A \leq 10^8$, every composite integer m satisfying $P(A + 1) < m \leq A + 1$ also satisfies inequality (20) for at least one prime divisor $q \mid m$, except precisely for*

$$(A, m) = (3, 4), (8, 9), (9, 9), (9, 10).$$

Computer-assisted proof. The accompanying directory `computational_reproducibility_files` contains `staircase_criterion.c`. This exact integer computation factors every relevant m and tests (20). It does not enumerate semigroups or compute gap progressions. The completed run checked 1,449,077,567 pairs and returned exactly the four displayed failures. Its output is recorded in `staircase_criterion_1e8.log` in the same directory. $\square$

The case $(3, 4)$ gives the genuine exception (12); the case $(8, 9)$ is covered by Theorem 4.1; and the two cases at $A = 9$ are covered by Theorem 4.5.

Computer-assisted theorem 6.2 (Prime staircase formula in the verified range). *For every integer A satisfying*

$$1 \leq A \leq 4 \cdot 10^{18}, \quad A \neq 3,$$

we have

$$\boxed{M(A) = A(P(A + 1) - 1)}.$$

Moreover, $M(3) = 7$.

Proof. Proposition 6.1 and the three special cases settle $A \leq 10^8$. Above that range, the computation of Oliveira e Silva, Herzog, and Pardi shows that prime gaps with lower prime below $4 \cdot 10^{18}$ are at most 1476 [18, 17]. Therefore, for composite $A + 1$ in the remaining interval, the backward prime distance satisfies

$$A + 1 - P(A + 1) \leq 1475 < \sqrt{P(A + 1)} - 2,$$

and Theorem 5.7 applies. If $A + 1$ is prime, the left side of (21) is zero, so the hypothesis also holds. $\square$

The finite range does not control the possible exceptions beyond its endpoint. To obtain an unconditional asymptotic statement, let p^+ denote the prime following p and define $g_p = p^+ - p$.

Theorem 6.3 (Density-one prime staircase). *For every $\varepsilon > 0$, the exceptional set satisfies*

$$\#\{A \leq x : M(A) \neq A(P(A + 1) - 1)\} = O_\varepsilon(x^{73/100+\varepsilon}).$$

In particular, the prime staircase formula holds on a set of natural density 1.

Proof. If (21) fails for $p = P(A + 1)$, then, apart from finitely many small primes, $g_p > \sqrt{p} - 1$. At most g_p potentially exceptional values of A lie in the gap following p . This observation gives

$$E(x) \ll \sum_{\substack{p \leq x+1 \\ g_p > \sqrt{p}-1}} g_p.$$

On a dyadic block $X < p \leq 2X$, and for X sufficiently large, $\sqrt{p} - 1 \geq \sqrt{X} - 1 \geq \frac{1}{2}\sqrt{X}$. Consequently, on the restricted set where $g_p > \sqrt{p} - 1$, we have

$$g_p \leq 2X^{-1/2} g_p^2.$$

It follows that

$$\sum_{\substack{X < p \leq 2X \\ g_p > \sqrt{p}-1}} g_p \ll X^{-1/2} \sum_{X < p \leq 2X} g_p^2.$$

Stadlmann's estimate

$$\sum_{p \leq Y} g_p^2 \ll_\varepsilon Y^{123/100+\varepsilon}$$

therefore gives $O_\varepsilon(X^{73/100+\varepsilon})$ on each block [19]. Dyadic summation proves the theorem. $\square$

This is a density-one conclusion: it allows a zero-density exceptional set and does not assert that the formula holds for every sufficiently large A .

Remark 6.4. If one instead uses Heath-Brown's classical mean-square estimate

$$\sum_{p \leq X} g_p^2 \ll_\varepsilon X^{23/18+\varepsilon},$$

then the same argument gives the weaker exceptional-set bound $O_\varepsilon(x^{7/9+\varepsilon})$ [9]. Stadlmann's more recent improvement supplies the sharper density-one estimate stated above.

The density-one theorem still permits infinitely many exceptions. Eventual validity follows from the following square-root-scale hypothesis.

Hypothesis 6.5 (Square-root gap hypothesis $H_{\sqrt{\cdot}}$). For all sufficiently large primes p , we have

$$g_p \leq \sqrt{p} - 1.$$

Conditional theorem 6.6. *Under Hypothesis 6.5, the formula*

$$M(A) = A(P(A+1) - 1)$$

holds for all sufficiently large A .

Proof. We set $p = P(A+1)$. If $A+1$ is composite, then $A+1 - p \leq g_p - 1 \leq \sqrt{p} - 2$; if it is prime, the left side is zero. Theorem 5.7 applies. $\square$

Several standard conjectures on prime gaps are much stronger than Hypothesis 6.5. For example, Cramér's conjectural bound $g_p = O((\log p)^2)$, as well as the usual prime-gap consequences of Firoozbakht's conjecture, imply Hypothesis 6.5 for all sufficiently large p [5, 11]. Therefore, either conjecture would imply the eventual prime-staircase formula through Theorem 5.7. Andrica's conjecture, by contrast, does not provide the constant required by Hypothesis 6.5 [1].

Independently of any prime-gap hypothesis, the quadratic upper bound is attained exactly when $A+1$ is prime.

Theorem 6.7. *For every $A \geq 1$, we have*

$$M(A) = A^2 \iff A+1 \text{ is prime.}$$

Proof. If $A+1$ is prime, Proposition 5.1 gives genus A^2 , and Corollary 2.9 gives the reverse inequality. Suppose $A+1$ is composite. If $m \leq A$, then $g \leq A(m-1) \leq A(A-1) < A^2$. If $m = A+1$, any prime divisor $q \mid m$ gives a strictly positive loss in (11). These exhaust the possible multiplicities. $\square$

Even when the exact staircase formula is not known, the distance to the preceding prime controls its deficit. We define

$$\Delta(A) = A^2 - M(A).$$

Theorem 6.8 (Unconditional deficit estimate). *For every $A \geq 1$, the deficit satisfies*

$$0 \leq \Delta(A) \leq A(A+1 - P(A+1)).$$

In addition, the pointwise estimate gives

$$\Delta(A) \ll A^{1.525}, \quad M(A) = A^2 + O(A^{1.525}) = (1 + o(1))A^2.$$

Proof. The first assertion combines $M(A) \leq A^2$ with Corollary 5.2. Baker, Harman, and Pintz prove

$$A+1 - P(A+1) \ll A^{0.525}$$

for sufficiently large A [2]. This is the relevant unconditional pointwise short-interval estimate. Substitution gives the deficit estimate, and division by A^2 gives the asymptotic equivalence. $\square$

At every $A \neq 3$ for which the prime staircase formula is proved, the normalized deficit is

$$\boxed{\frac{\Delta(A)}{A} = A + 1 - P(A + 1).} \quad (22)$$

Therefore, throughout the computer-verified range and on the density-one set of Theorem 6.3, the normalized deficit is exactly the backward prime distance. Outside the proved cases, (22) remains conjectural.

7 Genus realization and inverse plateaux

The prime construction can be varied without losing control of the AP length. Allowing the Kunz coordinates to range in a bounded band produces all intermediate genera needed below.

Lemma 7.1 (Bounded-band realization). *Let $p \leq A + 1$ be prime, let $1 \leq B \leq A$, and set $L_B = \lceil B/2 \rceil$. Every vector*

$$(k_1, \dots, k_{p-1}) \in \{L_B, L_B + 1, \dots, B\}^{p-1}$$

is a valid Kunz vector of multiplicity p , and its semigroup satisfies

$$\boxed{\text{AP}(S) \leq \max\{p - 1, B\} \leq A.}$$

Proof. For every target coordinate, we have $k_i + k_j \geq 2L_B \geq B \geq k_r$. Hence all unwrapped Kunz inequalities hold; the wrapped inequalities are weaker. If $p \nmid d$, every p consecutive terms contain a positive multiple of p , giving $\text{AP}_d(S) \leq p - 1$. If $p \mid d$, a progression remains in one residue class, which contains at most B gaps. $\square$

Proposition 7.2 (Bounded-band genera). *For every prime $p \leq A + 1$ and $1 \leq B \leq A$, each integer genus in the interval*

$$\left\lceil \frac{B}{2} \right\rceil (p - 1) \leq g \leq B(p - 1)$$

is realized by a semigroup with $\text{AP}(S) \leq A$.

Proof. Start with all $p - 1$ coordinates equal to L_B . Any integer increment from 0 to $(B - L_B)(p - 1)$ can be distributed among the coordinates, each with capacity $B - L_B$. It follows that every intermediate coordinate sum occurs, and the genus equals that sum. $\square$

The bounded bands overlap as their upper coordinate bound varies. Let $p = P(A + 1)$ and $n = p - 1$. For $1 \leq B \leq A$, put

$$I_B = \left[\left\lceil \frac{B}{2} \right\rceil n, Bn \right].$$

The intervals cover $[n, An]$: $I_1 = \{n\}$, and $\lceil (B + 1)/2 \rceil \leq B$ shows that consecutive intervals overlap.

Theorem 7.3 (Realization up to the prime staircase). *For every $A \geq 1$ and every integer g satisfying*

$$1 \leq g \leq A(P(A+1) - 1),$$

there exists a numerical semigroup S with

$$g(S) = g, \quad \text{AP}(S) \leq A.$$

Equivalently, the set of realized genera satisfies

$$\boxed{\{g(S) : \text{AP}(S) \leq A\} \supseteq \{1, \dots, A(P(A+1) - 1)\}}.$$

Proof. Proposition 7.2 and the interval coverage realize every genus in $[n, An]$. Since $n = p - 1 \leq A$, it remains only to realize $1 \leq g \leq A$. The ordinary semigroup $\{0, g+1, g+2, \dots\}$ has gap set $\{1, \dots, g\}$ and $\text{AP}(S) = g$. This proof is unconditional and does not use the prime staircase conjecture. $\square$

This realization result allows the extremal function to be inverted at every level where the staircase formula is known. We define

$$a(g) = \min\{\text{AP}(S) : g(S) = g\}, \quad A_M(g) = \min\{A : M(A) \geq g\}.$$

By definition, $A_M(g) \leq a(g)$. A global equality does not follow formally: if $M(A)$ were larger than the prime lower bound, the preceding theorem would not realize the intervening genera.

Corollary 7.4 (Inverse function at an exact staircase level). *Suppose that the exact staircase formula*

$$M(A) = A(P(A+1) - 1)$$

holds. It follows that every $1 \leq g \leq M(A)$ is realized with $\text{AP}(S) \leq A$. If

$$A_0 = \min\{A : M(A) \geq g\}$$

and the staircase formula is known at A_0 , then

$$\boxed{a(g) = A_0}.$$

Proof. Theorem 7.3 gives $a(g) \leq A_0$, and the definition of A_0 gives $A_0 \leq a(g)$. $\square$

Computer-assisted corollary 7.5 (Inverse staircase in the verified range). *For every $g \leq M(4 \cdot 10^{18})$, we have*

$$\boxed{a(g) = \min\{A : g \leq M(A)\}},$$

where in the relevant range

$$M(A) = \begin{cases} 7, & A = 3, \\ A(P(A+1) - 1), & A \neq 3. \end{cases}$$

Proof. The minimizing A is at most $4 \cdot 10^{18}$ by monotonicity. Apply Theorem 6.2 and Corollary 7.4; when $A = 3$, genera through 6 are covered by Theorem 7.3, and genus 7 by $\langle 4, 7, 13 \rangle$. $\square$

Exact consecutive values of M now translate directly into plateaux for the inverse function.

Corollary 7.6. *Suppose the staircase formula is known at A and $M(A - 1)$ is known exactly. Under these assumptions, the implication*

$$M(A - 1) < g \leq M(A) \implies \boxed{a(g) = A.}$$

holds. In particular, this implication gives

$$\boxed{a(g) = 10 \quad (55 \leq g \leq 100).}$$

Proof. The lower bound follows from the definition of $M(A - 1)$ and the upper bound from Corollary 7.4. The example uses $M(9) = 54$ and $M(10) = 100$; the latter follows from Theorem 6.7 because 11 is prime. $\square$

The plateau length is $M(A) - M(A - 1)$ whenever the two consecutive values and the realization at A are known. The set of levels at which the exact formula holds simultaneously at $A - 1$ and A has density one, since the exceptional set in Theorem 6.3 and its unit translate both have density zero. This density statement concerns staircase levels and does not by itself give a global closed formula for $a(g)$.

8 Further directions

The results above leave one main structural question unresolved: whether the prime staircase formula holds for every $A \neq 3$. The prime-gap input explains the finite verification and density-one results, but the harder remaining step lies within the semigroup problem itself. The arguments of Sections 5 and 6 exclude most multiplicities using the loss from a single divisor orbit; what remains is a narrow class of composite multiplicities for which this first loss is insufficient. Understanding how the different orbit constraints interact with the Kunz inequalities is the central task.

Two secondary directions concern settings in which the invariant may be useful beyond the extremal problem itself. Weierstrass semigroups and one-point algebraic-geometry codes provide one such setting, because the arrangement of the gaps already plays an important role. Another consists of special classes of numerical semigroups, such as symmetric semigroups, for which arithmetic progressions in the gap set can be related directly to arithmetic progressions among the nongaps.

To isolate the remaining structural obstruction, let $p = P(A + 1)$, and suppose that a composite multiplicity $m > p$ survives the single-divisor criterion of Theorem 5.7. Lemma 5.6 then forces

$$m - p > \sqrt{m} - 2.$$

Thus, a surviving multiplicity can occur only when $A + 1$ lies unusually far from the preceding prime. Moreover, the least-factor estimate is weakest when the least prime factor of m is close to $\sqrt{m}$. This singles out prime squares and balanced semiprimes as the most natural cases to understand.

The first example already occurs at $A = 9$. The multiplicities $9 = 3^2$ and $10 = 2 \cdot 5$ lie between the consecutive primes 7 and 11, and the single-divisor estimate does not recover the sharp bound. For $m = 9$, the additional information comes from anchoring. The Run Lemma controls not only the minimum Kunz coordinate on an orbit but also

the position at which that minimum first occurs. Once this positional information is combined with the Kunz inequalities, bounds propagate to further coordinates and recover the correct genus.

The same mechanism should persist more generally. For instance, when $m = q^2$, every nonzero q -orbit satisfies

$$\min_{i \in \mathcal{O}} k_i \leq \left\lfloor \frac{A}{q} \right\rfloor.$$

These small coordinates cannot be placed independently because the Kunz inequalities relate them through addition in $\mathbb{Z}/q^2\mathbb{Z}$. The problem is therefore not only to measure the loss on each orbit but also to understand how minima on different orbits force one another.

This suggests the following more precise question.

Problem 8.1 (Propagated loss). Find a quantitative genus-loss estimate for prime-square and balanced semiprime multiplicities that combines Run-Lemma anchoring with Kunz subadditivity, and that is strong enough to rule out the remaining composite competitors in the prime staircase conjecture.

Even a sufficiently uniform additional loss would already have consequences. For example, a bound of the form

$$g \leq A(m-1) - cAm$$

with some fixed $c > 0$ in the residual cases, together with $P(x) = (1 - o(1))x$, would eliminate such multiplicities for all sufficiently large A . A sharper analysis of the propagation phenomenon may therefore lead to a proof of the prime staircase conjecture without any hypothesis on prime gaps.

Beyond the extremal problem, Weierstrass semigroups provide a natural setting in which the arrangement of gaps matters. If P is a rational point on a smooth projective curve, then its Weierstrass semigroup $H(P)$ is numerical and its gap set has cardinality equal to the genus [20]. In the theory of one-point algebraic-geometry codes, the Feng–Rao order bound depends on additive relations among nongaps, while consecutive Weierstrass gaps are already known to influence distance estimates [20, 8]. It is therefore natural to ask whether the finer spectrum $\Sigma(H(P))$, rather than only $\text{AP}(H(P))$, carries useful information about the Feng–Rao sequence or the order bound. Structured Weierstrass semigroups, particularly those arising from recursive or otherwise highly organized families, provide natural test cases for this question.

Symmetric numerical semigroups provide another structured class. In this case, arithmetic progressions of gaps admit a simple reflection across the Frobenius number. Suppose that

$$x, x + d, \dots, x + (k-1)d \subseteq G(S).$$

Symmetry then gives the reflected progression

$$F - x - (k-1)d, \dots, F - x \subseteq S \cap [0, F].$$

Consequently, the part of the spectrum $\Sigma(S)$ arising from gap differences $d \in G(S) \setminus \{1\}$ may also be viewed through arithmetic progressions among the nongaps below F . It would be interesting to determine how much of the gap spectrum can be recovered from this reflected picture.

These considerations lead to four concrete questions.

1. Establish a propagated genus-loss theorem for the residual prime-square and balanced-semiprime multiplicities by combining Run-Lemma anchoring with Kunz subadditivity. Such a theorem is the main structural route to proving the prime staircase conjecture for every $A \neq 3$ without prime-gap hypotheses or finite computation.
2. Find a conceptual replacement for the remaining finite certificates, especially the multiplicities 9 and 10 at $A = 9$.
3. Determine whether $A = 3$ is the only exception to the prime staircase formula and, separately, classify the semigroups attaining $M(A)$. The value predicted by the conjecture does not imply uniqueness of a maximizing semigroup.
4. Understand the gap-difference part of $\Sigma(S)$, in particular for structured classes such as symmetric and Weierstrass semigroups.

The first question is the immediate continuation of the present work. The calculation at $A = 9$ suggests that the missing ingredient is a systematic way of propagating the positional information supplied by the Run Lemma through the Kunz inequalities.

A Computational certificates and reproducibility

All filenames in this appendix refer to files in the accompanying directory `computational_reproducibility_files`.

The finite search used in Proposition 4.4 was performed by `plateau_certificate.c`; its corresponding output is designated `plateau_certificate.log`. It enumerates the bounded Kunz boxes for $m = 8, 9, 10$ and uses only the Kunz inequalities and the explicitly stated Run-Lemma constraints. It does not construct full gap sets or compute $\text{AP}_d(S)$ directly, and it does not assert that these necessary constraints characterize the condition $\text{AP}(S) \leq 9$. A second exhaustive enumeration, implemented in `plateau_certificate_independent.c`, uses a different branching order and a disjunctive encoding of the orbit conditions. It produced the same maxima and the same unique maximizing vectors; its output is recorded in `plateau_certificate_independent.log`. Both implementations and their outputs accompany this draft.

As an independent check, the program `direct_ap_search.c` constructs gap sets and computes their arithmetic progressions directly. Its `a8` mode proves Theorem 4.1; the output is designated `direct_ap_A8.log`. Its `table` mode produces the comparison displayed after Conjecture 5.4; the output is designated `direct_ap_table_12.log`. Unlike the finite Kunz certificate, this program constructs each candidate gap set and computes the arithmetic progressions themselves. Its `extremizer9` mode directly checks the multiplicity-9 sharpness example, with output in `direct_ap_extremizer9.log`, while `tests` runs the elementary regression checks recorded in `direct_ap_tests.log`.

The exact inequality calculation in Proposition 6.1 was performed by `staircase_criterion.c`; the file `staircase_criterion_1e8.log` records the run through 10^8 . This calculation factors the relevant multiplicities and tests the exact composite criterion; it does not enumerate numerical semigroups. The companion `staircase_chunk.c` performs the same calculation on a specified interval, with a sample output in `staircase_chunk_sample.log`. A second complete execution on $1 \leq A \leq 10^8$ is recorded in `staircase_chunk_1e8.log`; it returns the same pair count and the same four failures as `staircase_criterion_1e8.log`.

For reproducibility, the direct search uses dynamically sized Kunz arrays and a dy-

namically allocated gap array indexed from 0 through the computed Frobenius number. An earlier fixed-width 128-bit mask overflowed when a possible gap position exceeded 127, producing a false discrepancy at $A = 12$; widening the representation removed it. Future extensions should allocate on the order of $A(A + 1)$ positions and avoid fixed Kunz arrays tied to a small multiplicity.

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Conflict of interest

The author declares no conflict of interest.

Code and data availability

The source code, complete computational logs, concise certificate, checksums, and reproduction instructions accompany this draft but are not yet deposited in a permanent public repository. The package contains the primary and independent plateau certificates, the direct AP checker, the two staircase-criterion programs, and their complete logs. The directory `computational_reproducibility_files` contains compilation and reproduction commands in `README_COMPUTATIONS.md`; the generated summary and checksums are `CERTIFICATE.txt` and `SHA256SUMS`. These files will be placed in a permanent repository before public circulation.

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